

MH2500 Probability and Introduction to Statistics

Slides for Week 12-2

Topics: LLN and CTL

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Chapter 8

We will discuss in this chapter two famous theorems in probability, the law of large numbers, and the central limit theorem.

- The law of large numbers, essentially says that when an experiment is repeated many times, the proportion of experiments with successful outcomes is approximately the probability of success.
- The central limit theorem quantifies how close the proportion is to the probability of success.

$$\{X \geq t\} = \{\omega \in \Omega : X(\omega) \geq t\}$$

Markov's inequality

Let X be a *nonnegative* random variable, then

$$P\{X \geq t\} \leq \frac{\mathbb{E}(X)}{t} \quad \text{for any } t > 0.$$

Proof: By

$$\mathbf{1}_{\{X \geq t\}}^{(\omega)} = \begin{cases} 1 & X^{(\omega)} \geq t \\ 0 & X^{(\omega)} < t \end{cases} \quad \begin{aligned} & \leftarrow \frac{X^{(\omega)}}{t} \geq 1 \\ & \leftarrow \frac{X^{(\omega)}}{t} < 1 \end{aligned}$$

$$\mathbf{1}_{\{X \geq t\}} \leq \frac{X}{t}, \quad \checkmark \quad \mathbf{1}_{\{X \geq t\}}^{(\omega)} \leq \frac{X^{(\omega)}}{t}$$

taking expectation on both sides, we have

$$P\{X \geq t\} = \mathbb{E}(\mathbf{1}_{\{X \geq t\}}) \leq \mathbb{E}\left(\frac{X}{t}\right) = \frac{\mathbb{E}(X)}{t}.$$

two sides
Exercise in the book
- one-side.

Chebyshev's Inequality

Let X be a random variable such that $\mathbb{E}(X)$ and $\text{Var}(X)$ exist. Then

$$P\{|X - \mathbb{E}(X)| \geq t\} \leq \frac{\text{Var}(X)}{t^2} \text{ for any } t > 0.$$

Proof:

$$P\{\omega \in \Omega : |X(\omega) - \mathbb{E}(X)| \geq t\}$$

$$Y = (X - \mathbb{E}(X))^2$$

$$P\{|X - \mathbb{E}(X)| \geq t\} = P\{|X - \mathbb{E}(X)|^2 \geq t^2\}$$

$$\leq \frac{\mathbb{E}(Y)}{t^2} \quad \leftarrow \text{By Markov's inequality}$$

$$\leq \frac{\mathbb{E}(|X - \mathbb{E}(X)|^2)}{t^2}$$

$$= \frac{\mathbb{E}(X - \mathbb{E}(X))^2}{t^2} \quad \leftarrow \text{Var}(X)$$

$$= \frac{\text{Var}(X)}{t^2}$$

$$P\{|X - E(X)| \geq t\} \leq \frac{\text{Var}(X)}{t^2}$$

$$P\{|X| \geq t\} \leq \frac{1}{t^2}$$

$$\boxed{k \cdot 1}$$

$$P\{|X| \geq k\} \leq \frac{1}{k^2}$$

||

$$P\{X \geq k\} + P\{X \leq -k\}$$

||

$$P\{X = k\} + P\{X = -k\}$$

||

$$\frac{1}{2k^2} + \frac{1}{2k^2}$$

||

$$\frac{1}{k^2}$$

Consequences

- Let $\sigma = \sqrt{\text{Var}(X)}$ and set $t = k\sigma$,

$$P\{|X - \mu| \geq k\sigma\} \leq \frac{\sigma^2}{(k\sigma)^2} = \frac{1}{k^2}.$$

Chebyshev's inequality

E.g., for $k = 2$, $P\{|X - \mu| \geq 2\sigma\} \leq \frac{1}{4}$.

- This is far from the tightness for many particular distributions: For a normal variable X , $P\{|X - \mu| \geq 2\sigma\} \approx 1/20$.
- But **Chebyshev's inequality is tight**. Consider X such that

$$P\{X = k\} = P\{X = -k\} = \frac{1}{2k^2}, \quad P\{X = 0\} = 1 - \frac{1}{k^2}.$$

$$\begin{aligned}
 P\{X = k\} &= P\{\omega \in \Omega : X(\omega) = k\} = \frac{1}{2k^2} \\
 P\{X = -k\} &= \frac{1}{2k^2} \\
 P\{X = 0\} &= 1 - \frac{1}{k^2} \\
 E(X) &= k \cdot \frac{1}{2k^2} + (-k) \cdot \frac{1}{2k^2} + 0 \cdot \left(1 - \frac{1}{k^2}\right) = 0 \\
 \text{Var}(X) &= E(X^2) - (E(X))^2 \\
 &= E(X^2) = 1
 \end{aligned}$$

Corollary

If $\text{Var}(X) = 0$, then $P\{X = \mu\} = 1$.

Proof: Observe that

$$\{|X - \mu| > 0\} = \bigcup_{n=1}^{\infty} \left\{ |X - \mu| \geq \frac{1}{n} \right\}.$$

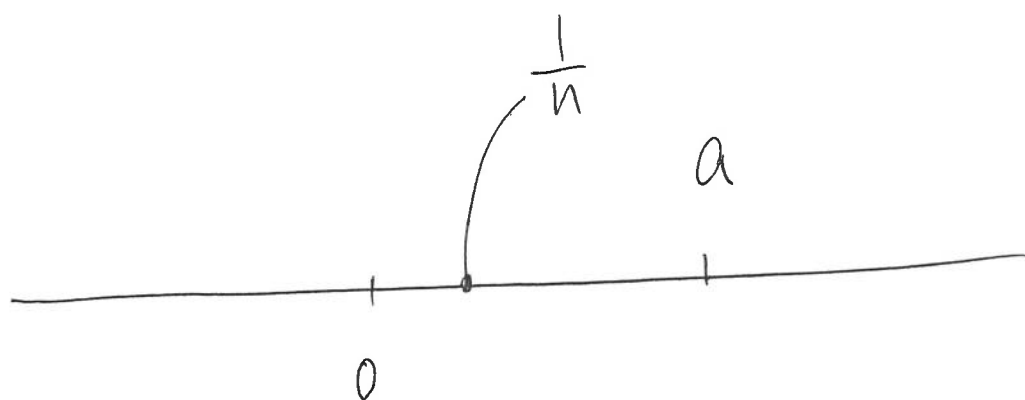
By Chebyshev's inequality, for any $\epsilon > 0$,

$$P\{|X - \mu| \geq \epsilon\} \leq \frac{\text{Var}(X)}{\epsilon^2} = 0.$$

Hence

$$P\{|X - \mu| > 0\} \leq \sum_{n=1}^{\infty} P\left\{ |X - \mu| \geq \frac{1}{n} \right\} = 0.$$

- How can we get the first inequality?



$a > 0 \iff$ there is some n such that

$$|X(\omega) - \mu| > 0 \iff \exists n \left(0 < \frac{1}{n} < |X(\omega) - \mu| \right)$$

↑
depends on ω

$$\begin{aligned} & \{\omega \in \Omega : |X(\omega) - \mu| > 0\} \\ &= \{\omega \in \Omega : \exists n (0 < \frac{1}{n} < |X(\omega) - \mu|)\} \\ &\subseteq \bigcup_{n \geq 1} \{\omega \in \Omega : \frac{1}{n} < |X(\omega) - \mu|\} \\ &\equiv \end{aligned}$$

The Law of Large Numbers

- 1 If we toss a fair coin many times, we expect about half of them are heads.
- 2 While detained as a prisoner during World War II, South African mathematician [John Kerrich](#) tossed a coin 10,000 times. He obtained 5067 heads.
- 3 Mathematically, the law of large numbers tells us that happens.
- 4 The coin tosses are random independent events. Let X_i be the indicator variable of obtaining a head on the i -th toss. Then the X_i 's are independent.

The proportion of heads is then given by $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. The law of large numbers tells us that $\bar{X}_n$ approaches $\frac{1}{2}$.

In the case of a fair coin toss, the X_i 's are Bernoulli random variables with $p = 1/2$,

$$\mathbb{E}(X_i) = \frac{1}{2} \quad \text{Var}(X_i) = \frac{1}{4},$$

and for $n = 10000$,

$$\bar{X}_{10000} = \frac{1}{10000} (X_1 + \dots + X_{10000})$$

$$\text{Var}(\bar{X}_{10,000}) = 2.5 \times 10^{-5}.$$

Hence the standard deviation is 0.005. $\frac{1}{(10000)^2} \cdot (\text{Var}(X_1) + \dots + \text{Var}(X_{10000})) = \frac{1}{10000} \times \frac{1}{4} = 0.25 \times 10^{-4} = 2.5 \times 10^{-5}$

Kerrich's observation was 0.5067. This is 0.0067 away from the mean, which is slightly more than one standard deviation away.

This is consistent with Chebyshev's inequality in the form

$$P\{|\bar{X}_n - \mu| > k\sigma\} \leq 1/k^2.$$

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

$$= \frac{1}{n} \text{Var}(X_i)$$

$$= \frac{1}{n} \times \frac{1}{4}$$

When $n=100000$

$$\frac{1}{n} \times \frac{1}{4} = 0.25 \times 10^{-4}$$

$$= 2.5 \times 10^{-5}$$

Weak Law of Large Numbers

Let $X_1, X_2, \dots, X_i, \dots$ be a sequence of i.i.d. variables with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P\{|\bar{X}_n - \mu| > \epsilon\} = 0.$$

Proof: We have calculated before that

$$\mathbb{E}(\bar{X}_n) = \mu \quad \text{and} \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The desired result now follows immediately from Chebyshev's inequality, which states that

$$P\{|\bar{X}_n - \mu| > \epsilon\} \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$$

when $n \rightarrow \infty$.

Weak because the probability of “large deviation from mean” goes to 0, but large deviation can still happen.

In fact, a stronger statement holds: “Large deviation” can never happen once n is large enough.

Strong Law of Large Numbers

Let $X_1, X_2, \dots, X_i, \dots$ be a sequence of i.i.d. variables with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then

$$P \left\{ \lim_{n \rightarrow \infty} \bar{X}_n = \mu \right\} = 1.$$

$\equiv P \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \bar{X}_n(\omega) = \mu \right\}$
 $\equiv \lim_{n \rightarrow \infty} P \left(\frac{X_1(\omega) + \dots + X_n(\omega)}{n} = \mu \right)$

This followed from the fact that

$$\text{Var} \left(\frac{S_n}{n} \right) = \frac{1}{n^2} \text{Var}(S_n) = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n} \rightarrow 0.$$

The law of large numbers says that S_n/n converges to μ in probability.

Central Limit Theorem

The central limit theorem is concerned not with the fact that the ratio S_n/n converges to μ but with **how it fluctuates around μ** .

- Normalizing the S_n 's, set

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

The central limit theorem states that Z_n converges to the standard normal.

Central Limit Theorem

Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables having mean μ and variance σ^2 .

- Also assume that their moment-generating function M is defined in a neighbourhood of zero.

Let

$$S_n = \sum_{i=1}^n X_i.$$

Then

$$\lim_{n \rightarrow \infty} P \left\{ \frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x \right\} = \Phi(x), \quad -\infty < x < \infty.$$

Central Limit Theorem – Example 1

If 10 fair dice are rolled, find the approximate probability that the sum obtained is between 30 and 40, inclusive.

Solution: Let X_i denote the value of the i -th die.

$$\mathbb{E}(X_i) = \frac{7}{2}, \quad \text{Var}(X_i) = \frac{35}{12}.$$

Let $X = X_1 + \cdots + X_{10}$.

$$\begin{aligned} P\{30 \leq X \leq 40\} &= P\{29.5 \leq X \leq 40.5\} \\ &= P\left\{ \frac{29.5 - 35}{\sqrt{350/12}} \leq \frac{X - 35}{\sqrt{350/12}} \leq \frac{40.5 - 35}{\sqrt{350/12}} \right\} \\ &\approx P\{-1.0184 \leq Z \leq 1.0184\} \\ &= 2\Phi(1.0184) - 1 \approx 0.692 \end{aligned}$$

Here Z is the standard normal distribution.

Central Limit Theorem – Example 2

Let $X_1, \dots, X_{10}$ be i.i.d. uniform random variables on $(0, 1)$.

$$\text{Find } P\left\{\sum_{i=1}^{10} X_i > 6\right\}.$$

Solution: As each X_i is a uniform random variables on $(0, 1)$, we have

$$\mathbb{E}(X_i) = \frac{1}{2} \text{ and } \text{Var}(X_i) = \frac{1}{12}.$$

• Let $X = X_1 + \dots + X_{10}$. Then

$$P\{X > 6\} = P\left\{\frac{X - 5}{\sqrt{10/12}} > \frac{6 - 5}{\sqrt{10/12}}\right\}$$

$$\approx P\{Z > 1.095\} = 1 - \Phi(1.095) \approx 0.1367$$

Here Z is the standard normal distribution.

Continuity Theorem

- Recall that a distribution function is **uniquely determined** by its mgf M_n .
- The following theorem (**proof omitted**) states that this unique determination holds for limits as well.

Continuity Theorem

- Let F_n be a sequence of cdf with M_n the corresponding moment-generating function.
- Let F be a cdf with the moment-generating function M .

If $M_n(t) \rightarrow M(t)$ for all t in an open interval containing zero, then $F_n(x) \rightarrow F(x)$ at all continuity points of F .

Proof of CTL:

We first assume that $\mu = 0$ and $\sigma = 1$.

- Since the CDF of $N(0, 1)$ is continuous, it suffices to show that the mgf of $(X_1 + \dots + X_n)/\sqrt{n}$ converges to the mgf of $N(0, 1)$.

Assume that $M(t)$, the mgf of X_i , exists and is finite. Then

$$M\left(\frac{t}{\sqrt{n}}\right) = \mathbb{E}\left(e^{\frac{t}{\sqrt{n}}X_i}\right) = M\left(\frac{t}{\sqrt{n}}\right) \cdot M\left(\frac{t}{\sqrt{n}}\right) \cdots M\left(\frac{t}{\sqrt{n}}\right)$$

$\swarrow \frac{X_1}{\sqrt{n}} \quad \swarrow \frac{X_2}{\sqrt{n}} \quad \swarrow \frac{X_n}{\sqrt{n}}$

The mgf of $(X_1 + \dots + X_n)/\sqrt{n}$ is $\left[M\left(\frac{t}{\sqrt{n}}\right)\right]^n$. Let $L(t) = \log M(t)$.

Then $L'(t) = \frac{M'(t)}{M(t)}$ and $L''(t) = \frac{M(t)M''(t) - [M'(t)]^2}{[M(t)]^2}$, which gives

- $L(0) = 0$, $L'(0) = M'(0) = \mu = 0$, $L''(0) = M''(0) = \mathbb{E}(X^2) = 1$.
- $\underbrace{L(0)}_{=\log(1)} = 0$ $\underbrace{L'(0)}_{=\frac{M'(0)}{M(0)} = 0} = 0$

mgf of $\frac{X_1 + \dots + X_n}{n}$

mgf for $N(0,1)$

To prove $\lim_{n \rightarrow \infty} [M(t/\sqrt{n})]^n \rightarrow e^{t^2/2}$, we just need to show:

$$\lim_{n \rightarrow \infty} [n \log M(t/\sqrt{n})] \rightarrow t^2/2,$$

i.e.,

$$\lim_{n \rightarrow \infty} [nL(t/\sqrt{n})] \rightarrow t^2/2,$$

which is true:

$$\begin{aligned} \lim_{n \rightarrow \infty} nL(t/\sqrt{n}) &= \lim_{n \rightarrow \infty} \frac{L(t/\sqrt{n})}{n^{-1}} = \lim_{n \rightarrow \infty} \frac{-L'(t/\sqrt{n})n^{-3/2}t}{-2n^{-2}} \\ &= \lim_{n \rightarrow \infty} \left[\frac{L'(t/\sqrt{n})t}{2n^{-1/2}} \right] = \lim_{n \rightarrow \infty} \left[\frac{-L''(t/\sqrt{n})n^{-3/2}t^2}{-2n^{-3/2}} \right] \\ &= \lim_{n \rightarrow \infty} \left[L''\left(\frac{t}{\sqrt{n}}\right) \frac{t^2}{2} \right] = \frac{t^2}{2}. \end{aligned}$$

In general,

If $\mu \neq 0$, or $\sigma^2 \neq 1$, then take $Y_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}(Y_i) = 0$ and $\text{Var}(Y_i) = 1$.

By what proved just now, we have

$$\lim_{n \rightarrow \infty} P \left\{ \frac{Y_1 + \dots + Y_n}{\sqrt{n}} \leq x \right\} = \underline{\underline{\Phi(x)}}, \quad -\infty < x < \infty,$$

which is what wanted

$$\lim_{n \rightarrow \infty} P \left\{ \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq x \right\} = \underline{\underline{\Phi(x)}}.$$

Remarks:

- 1 We have given one of the simplest version of the central limit theorem. (There are many other versions.)
- 2 We assumed the mgf exists, which is a rather strong assumption.